Oxygenation versus composition in positronium-lifetime imaging: a quantified identifiability limit and a lipid-driven hypoxia bias

Author: Ernest Darell Zermeño[1,2]

[1] Universidad Panamericana, Mexico. [2] Universidad de Guadalajara, Guadalajara, Mexico. Correspondence: 0244552@up.edu.mx.

Preprint — modeling / methods. Version 1.

Abstract

Background. Ortho-positronium (o-Ps) lifetime, now measurable in vivo on long-axial-field-of-view (LAFOV) PET/CT, has been proposed as a biomarker of tissue oxygenation and hypoxia. The o-Ps lifetime, however, depends on **both** dissolved oxygen and tissue composition (free-volume/lipid content). That these are confounded is widely acknowledged qualitatively, but, to our knowledge, the confound has not been quantified as an oxygen-versus-composition identifiability/CRLB problem.

Aim. To quantify (i) whether pO_2 and composition are separable from o-Ps observables, (ii) the hypoxia-imaging bias that follows from ignoring composition, and (iii) the statistical conditions under which separation is — or is not — feasible.

Methods. We build a transparent rate-additivity forward model mapping (pO_2, lipid fraction f) to the o-Ps lifetime τ_3 and intensity I_3, with constants sourced and the two load-bearing unknowns flagged. We derive the naive-reader bias, set up the Fisher-information / Cramér–Rao analysis for $\theta = (pO_2, f)$ from τ_3 alone and from (τ_3, I_3), compute a rigorous Poisson-spectrum Fisher information (with spectral-nuisance profiling), validate fixed-nuisance CRLB attainability by parametric Monte Carlo, and translate the bound into clinically meaningful counts and ROI volumes for realistic isotopes.

Results. From the lifetime alone, pO_2 and composition are **structurally non-identifiable** (rank-1 Fisher). Even 2–5% lipid throws a naive hypoxia read by hundreds of mmHg, so an uncorrected lifetime map is composition-dominated. A second observable — the o-Ps intensity I_3 — restores identifiability **in principle**, but weakly: the Cramér–Rao bound on pO_2 is about 330–760 mmHg at 10^5 o-Ps counts (a bracket spanning optimistic to nuisance-profiled noise models). A parametric Monte Carlo confirms that the fixed-nuisance Poisson bound is statistically attainable under the assumed spectrum model. Clinically useful separation ($\sigma_{pO_2} \leq 10$ mmHg) requires **litres** of pooled signal even with ^{124}I, and is impossible at native voxel resolution. We also correct a planning-stage error the model itself exposed: the separability condition is a non-collinear Jacobian; in the pessimistic case where I_3 has no oxygen response, the composition handle is $\partial I_3/\partial(\text{composition}) \neq 0$.

Conclusion. Positronium-lifetime hypoxia mapping is identifiability- and statistics-limited. We provide the bias formula, the count/volume/isotope requirements, the second-observable prescription, and the decisive bench experiment that would settle the two unmeasured constants.

Keywords: positronium lifetime imaging; ortho-positronium; tissue oxygenation; hypoxia biomarker; identifiability; Cramér–Rao bound; PET/CT.

---

1. Introduction

In a fraction of positron–electron annihilations the pair first forms positronium; its triplet form, ortho-positronium (o-Ps), is metastable and in matter localises in nanometre-scale free-volume voids, annihilating predominantly by pick-off. Its lifetime ($\approx$ 1.5–3 ns in soft tissue) therefore reports the size of those voids — i.e. tissue composition and microstructure. Paramagnetic molecular oxygen additionally shortens the o-Ps lifetime through spin (ortho$\rightarrow$para) conversion, with a rate linear in dissolved-O_2 concentration [1,2]. This oxygen sensitivity is the physical basis for proposing o-Ps lifetime as an in-vivo imaging biomarker of tissue oxygenation and hypoxia [1,3,4] — a clinically valuable target, since hypoxic tumour regions are radio- and chemo-resistant and current surrogates (e.g. ^{18}F-FMISO PET) are limited.

Positronium lifetime imaging (PLI) has moved from the bench to the clinic: dedicated multi-photon systems and commercial LAFOV PET/CT scanners now produce in-vivo human positronium images and per-region lifetimes [5,6]. These are real achievements. They do not, by themselves, establish that an in-vivo lifetime *contrast* reflects oxygenation, because the lifetime is dominated by composition: between soft tissues the o-Ps lifetime varies by about 0.5 ns (adipose 2.54 ns vs muscle/hepatic about 2.0 ns [7]), whereas the oxygen-specific term implied by published quenching constants is on the order of a few picoseconds across physiological dissolved-oxygen differences [1]. The oxygen signal therefore sits inside a composition baseline two orders of magnitude larger, and — tellingly — published oxygen calibrations central to these claims have been performed in water or aqueous samples [1], where composition is fixed and the confound is invisible by design.

The field acknowledges this confound qualitatively: o-Ps lifetime is "the convolution of several phenomena, such as oxygen and free-radical concentration, in addition to the average void radius" [7], and reviews note that it is "presently not known how much o-Ps lifetime increases when pO_2 decreases by 50 mmHg" in living tissue. But no work has written down what that does to *identifiability* or to the resulting *imaging bias*. That is the gap this paper fills.

Recent work has proposed joint lifetime and $3\gamma/2\gamma$ readouts for pO_2 [12,13,16]. Here we make the composition problem quantitative. We (i) prove that pO_2 and composition are structurally non-identifiable from the lifetime alone; (ii) quantify the lipid-driven hypoxia bias in mmHg; (iii) determine when a second observable (the o-Ps intensity I_3, or the $3\gamma/2\gamma$ ratio) breaks the degeneracy, and at what statistical cost; (iv) translate the bound into count, ROI-volume and isotope requirements under explicit yield scenarios; and (v) specify the bench experiment that would measure the two load-bearing quantities. This is a *limits* paper, not a claim that o-Ps senses oxygen. It is the in-silico complement to a companion single-subject in-vivo demonstration of the same confound in human cardiac PLI [17].

Positioning against prior art. The physical ingredients are established; the identifiability framing is not.

Line of prior work	What it established	Not addressed
O_2 quenching & mechanism — Shibuya [1], Stepanov [2]	o-Ps lifetime responds to dissolved O_2 via spin conversion; aqueous pO_2 calibration	calibrated at fixed composition, so the confound is invisible
Composition sensitivity — Avachat [7]	o-Ps lifetime discriminates soft tissues by free-volume/lipid	oxygenation; joint identifiability
In-vivo PLI — Moskal [3–5], Mercolli [6,11]	human positronium images and regional lifetimes (LAFOV / multi-photon)	whether a lifetime contrast reflects pO_2 or composition
Lifetime estimation — Berens [8], Köllner–Wolfrum [15]	photon economy for estimating a single lifetime	two-parameter (pO_2, f) separation
Multi-observable oxygen — Alkhorayef [12,13], Moskal [16]	$3\gamma/2\gamma$ and combined readouts proposed for oxygenation	the identifiability condition that makes a second observable necessary
Identifiability methodology — Sourbron–Buckley [14]	multi-parameter CRLB identifiability (DCE-MRI)	positronium
This work	**pO_2-vs-composition identifiability/CRLB; bias in mmHg; count/volume/isotope limits**	—

2. Methods

2.1 Rate-additivity forward model

We model the o-Ps decay rate as a sum of independent channels,

$$\lambda_3 = 1/\tau_3 = \lambda_{\text{pickoff}}(f) + \kappa(f)\,\alpha(f)\,pO_2,$$

where $f \in [0,1]$ is the lipid (free-volume-rich) fraction of a two-compartment tissue, $\lambda_{\text{pickoff}}(f)$ is the composition-dependent baseline rate, κ is the o-Ps O_2 quenching constant, and α the O_2 solubility (so that $[O_2] = \alpha\,pO_2$). The baseline lifetime $\tau_0(f)$ interpolates between water and lipid endpoints; the O_2 sensitivity is $|d\tau_3/dpO_2| = \tau_0^2\,\kappa\alpha$. The lipid:water oxygen-solubility ratio was anchored to published oil/lipid measurements [9,10]. The o-Ps intensity is modelled as composition-dependent, $I_3 = I_3^w + (\partial I_3/\partial f)\,f + (\partial I_3/\partial pO_2)\,pO_2$. Constants and sources are in Table 1. Two quantities are unmeasured in the biological lipid/protein environments relevant here and are treated as unknowns: the quenching constant in lipid, κ_{lipid} (swept over 0.1–1.0× the water value), and the composition slope $\partial I_3/\partial f$.

2.2 The naive-reader bias

A “naive reader” assumes pure-water calibration ($f = 0$) and inverts an observed lifetime to a pO_2. The apparent-pO_2 error as a function of true (pO_2, f) is the lipid-O_2 bias; because the composition-induced baseline shift dwarfs the oxygen term, this is the dominant error in any uncorrected hypoxia read.

2.3 Identifiability: structural and practical

For $\theta = (pO_2, f)$ we compute the Fisher information matrix (FIM) and the Cramér–Rao bound $\mathrm{Cov}(\hat{\theta}) \succeq F^{-1}$. **Structural** non-identifiability is read from the rank of F (a rank-deficient $F \Rightarrow$ infinite CRLB on a parameter combination); **practical** identifiability from the CRLB magnitudes and the parameter correlation ρ. This multi-parameter framing follows the same logic used in established imaging identifiability problems, for example dynamic contrast-enhanced MRI [14]. We evaluate the FIM (a) from the lifetime alone and (b) from (τ_3, I_3). Because from τ_3 alone both parameters enter through the single scalar λ_3, the two score vectors are collinear and F is rank-1 — the exact, structural source of the degeneracy. Explicitly, with the single score row $J = [\,\partial\lambda_3/\partial pO_2,\ \partial\lambda_3/\partial f\,]$ and lifetime-rate variance σ_λ^2,

$$F_\tau = \sigma_\lambda^{-2}\,J^\top J, \qquad \det F_\tau = 0,$$

so the CRLB on either parameter alone is infinite; only the scalar combination λ_3 is constrained.

2.4 Rigorous Poisson-spectrum FIM

The idealized two-observable FIM treats τ_3 and I_3 as clean measurements; the realistic case fits the whole o-Ps lifetime spectrum. We therefore build the spectral density $m(t\,|\,\theta)$ — a two-component exponentially-modified-Gaussian (a fast component plus the o-Ps component, whose slope is $\tau_3(\theta)$ and amplitude $I_3(\theta)$) on a large flat accidental background — and compute the per-bin Poisson FIM, $F_{ab} = \sum_k m_k^{-1}\,\partial_a m_k\,\partial_b m_k$. This includes automatically the τ_3–I_3 estimation covariance and is complementary to moment-based lifetime estimators and photon-economy analyses for lifetime measurements [8,15]. We additionally **profile** the spectral nuisances (background level, fast-component lifetime, prompt offset, instrument resolution) by extending the FIM to those parameters and marginalizing; the instrument response (σ_{IRF}, t_0) is normally calibrated from the prompt peak and is kept fixed in the realistic case. As an internal estimator check, we simulate Poisson spectra under the same forward model and fit (pO_2, f) by maximum likelihood to verify that the fixed-nuisance CRLB is attainable when the model is correctly specified.

2.5 Counts model and the separability boundary

We anchor the o-Ps triple-coincidence yield to a public ^{44}Sc PLI phantom preprint (about 3.6×10^4 o-Ps counts in a 5.57 mL ROI [11]) and scale counts with ROI volume and an effective isotope-yield factor (prompt branching and source-to-background; ^{124}I best in the scenario model, high-positron-energy ^{82}Rb worst). These isotope factors are scenario-level effective-yield scalings, not universal measured constants. For each tissue state we then compute the ROI volume required to reach a target σ_{pO_2}, and we classify a virtual tumour voxel-by-voxel as viable / marginal / impossible.

Software: Python 3.13, NumPy, SciPy, Matplotlib; all code, constants and figures are released (Code availability).

3. Results

3.1 The lifetime alone cannot separate oxygenation from composition

Because $\mathrm{pO_2}$ and f enter only through the single summed rate λ_3, the 2×2 Fisher information from τ_3 alone is rank-1: its determinant is zero and the CRLB on either parameter individually is infinite (Figure 4). This is **structural** non-identifiability — not a statistical shortfall but an algebraic one. Only the linear combination λ_3 is constrained.

3.2 A lipid-driven hypoxia bias of hundreds of mmHg

The composition baseline shift dominates the tiny oxygen term, so a naive water-calibrated read is catastrophically biased (Figure 1). For a tissue truly at $\mathrm{pO_2}$ = 20 mmHg, ignoring composition produces an apparent-$\mathrm{pO_2}$ error of −178 mmHg at 2% lipid, −439 mmHg at 5%, and −860 mmHg at 10% — values that are not merely large but unphysical (negative), showing the read is composition-dominated rather than oxygen-driven. These magnitudes are stated under the endpoint-mixing forward model, but the conclusion is robust to it: the bias scales approximately linearly with the composition-to-lifetime coupling (within about 0.5% over this sweep), so halving that coupling still leaves −222 mmHg at 5% lipid, and the bias falls below a clinically tolerable 10 mmHg only if the coupling is about 44× smaller than the measured adipose–water baseline gap (about 18 ps versus about 785 ps) — implausible for real soft tissues (a strongly convex $\tau_0(f)$ would be the main functional form left untested here). The O_2 sensitivity itself is composition-dependent, uncertain, and temperature-dependent: at body temperature it is about 0.09 ps/mmHg in water (O_2 solubility about 1.3 µM/mmHg at 37 °C), whereas the aqueous calibration of Shibuya et al. near room temperature (about 20 °C) implies about 0.13–0.14 ps/mmHg, the difference reflecting the temperature dependence of O_2 solubility; the body-temperature value is the one relevant in vivo and, being smaller, only strengthens the composition-dominance conclusion. In pure lipid the sensitivity rises to between about 0.09 and 0.90 ps/mmHg depending on the unmeasured κ_{lipid} (Figure 2) — the approximately 1–10× net sensitivity spread is precisely that unknown.

3.3 The intensity I_3 restores identifiability — and a corrected linchpin

In the pessimistic baseline model, oxygen acts only on the rate λ_3 and leaves the o-Ps formation probability I_3 unchanged, whereas composition moves both τ_3 and I_3. With the two-vector observable (τ_3, I_3) the score directions are no longer collinear and the FIM becomes full-rank: pO_2 and f are identifiable in principle. The formal condition is that the Jacobian of (λ_3, I_3) with respect to (pO_2, f) be non-collinear. The analysis corrected a planning-stage assumption: in the pessimistic case $\partial I_3/\partial pO_2 = 0$, the necessary composition handle is $\partial I_3/\partial f \neq 0$; as this slope $\to 0$ the CRLB diverges and the system is degenerate again. A non-zero $\partial I_3/\partial pO_2$ can help if it supplies a non-collinear oxygen-sensitive direction; it is not itself required. The dependence on this linchpin slope is quantified on the rigorous spectrum model ($pO_2 = 20$ mmHg, $f = 0.10$, 10^5 o-Ps counts): as $\partial I_3/\partial f$ shrinks, the CRLB on pO_2 diverges.

$\partial I_3/\partial f$	$\sigma(pO_2)$ (mmHg)
0.150 (nominal)	332
0.050	473
0.015	998
0	degenerate (∞)

3.4 The rescue is real but weak: the CRLB bracket

Identifiability does not imply precision. At 10^5 o-Ps counts the CRLB on pO_2 at a hypoxic fatty voxel ($pO_2 = 20$, $f = 0.10$) is, across noise models (Table 2): 332 mmHg (Poisson, nuisances fixed; optimistic), 570 mmHg (idealized analytic with the field's empirical lifetime precision), 665 mmHg (Poisson with the per-spectrum nuisances profiled; realistic), and 758 mmHg (all nuisances profiled;

pessimistic). The realistic figure is about 2× the optimistic one — the price of fitting the background and fast component. The τ_3–I_3 estimation covariance that this entails raises the parameter correlation from $\rho = 0.05$ (idealized) to $\rho = 0.70$ (spectrum). The conclusion is robust to the noise model: the bound is hundreds of mmHg at 10^5 o-Ps counts, implying about 3–5×10^8 o-Ps events for a 10 mmHg target in the profiled models. The spectrum FIM itself is parameterized by total in-window events, with $N_3 = I_3(1 - \text{background})N_{\text{total}}$; at the representative state, 10^5 o-Ps counts correspond to about 1.1×10^6 in-window events.

The fixed-nuisance Poisson CRLB is not only formal. In parametric Monte Carlo at $pO_2 = 20$ mmHg and $f = 0.10$, the maximum-likelihood estimator recovered the truth with empirical standard deviations matching the CRLB at both tested count levels (empirical/CRLB = 1.00–1.01 for pO_2; 0.98–1.01 for f). The pO_2 pull distribution was close to $N(0, 1)$ (mean +0.05, standard deviation 1.00; Supplementary Figure S1). This validates estimator attainability under the assumed spectrum model; it does not validate the unmeasured biological constants.

3.5 Feasibility: litres of pooling, voxel-scale impossible

Translated to the effective-yield scenarios (Figure 5, Figure 6), separating pO_2 from composition to $\sigma(pO_2) \leq 10$ mmHg requires ROI volumes of **litres**. At $pO_2 = 10$ mmHg and 5% lipid, the required pooled volume is about 20 L for the best-yield scenario (^{124}I), about 78 L for ^{68}Ga, and about 130 L for ^{82}Rb. Even at 60% lipid, where the higher O_2 sensitivity helps, the requirement remains about 5 L for ^{124}I. At a native 4 mm clinical voxel (0.064 mL), the separation is impossible in 100% of the virtual tumour (Figure 6). Paradoxically, fatty tissue is "easier" to separate (its O_2 sensitivity is higher) but is also the most biased — both consequences of the same lipid-O_2 coupling.

These volumes use the central lipid-quenching scenario ($\kappa_{\text{lipid}} = 0.3\times$ water); because κ_{lipid} is unmeasured, we stress-test it upward. Reported o-Ps sensitivities in organic liquids motivate an upper bound well above the water value — an upper stress test, not a biological claim for adipose tissue. Fatty-tissue feasibility is sensitive to this assumption; the lean, tumour-hypoxia regime (the clinically relevant one) is not:

$\kappa_{\text{lipid}}/\kappa_{\text{water}}$	ROI for $\sigma(pO_2) \leq 10$ mmHg, $f = 0.60$	$f = 0.05$
0.3 (central)	5.1 L	19.6 L
1.0 (water ceiling)	0.75 L	14.3 L
3.0 (stress)	98 mL	7.1 L
10 (organic-liquid stress)	9.5 mL	1.7 L

(^{124}I scenario.) The voxel-scale impossibility and the structural non-identifiability are unchanged by widening κ_{lipid}; only the fatty-tissue volume moves.

4. Discussion

The practical implication is methodological and stark: an o-Ps lifetime map is **not** a hypoxia map. Reading hypoxia from τ_3 alone is structurally impossible; doing it from (τ_3, I_3) is statistically out of reach at native voxel resolution and current yields by five to six orders of magnitude (three to four once pooled to millilitre-scale ROIs). This does not refute the physics — oxygen does shorten o-Ps lifetime — but it bounds what can be inferred and tells the field exactly how far feasibility lies and what would change it.

Two constructive prescriptions follow. **First**, oxygenation work must extract and report a second observable — the o-Ps **intensity** I_3 and/or the $3\gamma/2\gamma$ ratio — not the lifetime alone. This is consistent with recent proposals to combine lifetime and $3\gamma/2\gamma$ oxygen readouts [16], but here the requirement comes from the oxygen-versus-composition identifiability condition. It is the o-Ps analogue of adding a second echo to make a fat/water separation full-rank. **Second**, where a co-registered CT or MRI provides composition, the bias formula of §3.2 supplies a composition-aware correction to any future lifetime-based read. The boundary of §3.5 also gives the field an honest target for acquisition design (counts, ROI pooling, isotope) tuned to *separation* rather than to resolving a single lifetime.

The same degeneracy was observed in vivo in a companion single-subject human cardiac PLI re-analysis, where a reproducible right>left ventricular lifetime contrast did not survive a structure-matched control and was traced to composition and instrument behaviour rather than oxygenation [17]. The present model explains *why* such contrasts are composition-dominated and quantifies the limit.

The honest ceiling: this is a model-based analysis. Its force is conditional on the rate-additivity forward model and on two constants not yet measured in the biological lipid/protein environments of interest — κ_{lipid} and $\partial I_3/\partial f$. We have made both explicit and swept the first; the second is the separability handle in the pessimistic case where oxygen does not move I_3. The analysis therefore does not "solve" hypoxia imaging — it organizes the problem, bounds it under stated assumptions, and points to the decisive measurement.

5. The decisive experiment

The two load-bearing unknowns are settled by a single bench measurement: a positron-annihilation-lifetime (PALS) matrix over $O_2 \times$ {water, lipid emulsion, protein/serum}, at controlled dissolved-oxygen with an independent pO_2 ground truth, recording both τ_3 and I_3. This yields (i) κ in lipid and protein (fixing the bias magnitude and the sensitivity), and (ii) $\partial I_3/\partial(\text{composition})$ and $\partial I_3/\partial pO_2$

(fixing whether, and how strongly, the degeneracy can be broken). It is self-contained, requires no scanner, and would convert this limits analysis into a measured one.

6. Limitations

The forward model is a transparent two-compartment rate-additivity form; richer free-volume models (Tao–Eldrup) and intensity chemistry (inhibition, scavenging) are not included. The o-Ps intensity values are illustrative placeholders; the composition slope $\partial I_3/\partial f$ is assumed non-zero in the baseline scenario (its measurement is the experiment of §5). κ_{lipid} is unmeasured in the relevant biological lipid/protein environments and swept. The counts model is anchored to one public preprint yield and uses scenario-level isotope factors rather than universal measured constants. Nuisance profiling bounds the realistic CRLB but the spectrum model is itself idealized. In particular, I_3 here is the intrinsic o-Ps formation probability, whereas a scanner estimates a fitted o-Ps amplitude that also absorbs detection efficiency, scatter, attenuation, geometric acceptance, accidental background and prompt-gamma contamination; the FIM profiles the per-spectrum nuisances (background, fast component, IRF) but not this full imaging chain, so the realistic I_3-estimation covariance — and hence the CRLB — is if anything a lower bound on the in-system value. The Monte Carlo validates statistical attainability of the fixed-nuisance CRLB under the assumed forward spectrum; it is not a physical validation of κ_{lipid}, $\partial I_3/\partial f$, or $\partial I_3/\partial pO_2$. A minimal PALS measurement or a real-data I_3 re-analysis remains the natural empirical anchor.

7. Conclusion

From the positronium lifetime alone, tissue oxygenation and composition are structurally non-identifiable; ignoring composition biases a hypoxia read by hundreds of mmHg; the o-Ps intensity can break the degeneracy in principle but only with litres of pooled signal under current yield scenarios, making voxel-scale clinical pO_2 mapping infeasible without a 10^5–$10^6\times$ gain in effective o-Ps yield (a 10^3–$10^4\times$ gain would suffice only for about 5 mL pooled ROIs, not native voxels). We provide the bias formula, the identifiability and count/volume/isotope requirements, the corrected separability condition (a non-collinear second-observable Jacobian), and the decisive experiment. Positronium-lifetime contrasts should not be interpreted as oxygenation maps until composition is controlled and a second observable is measured.

Declarations

Ethics. This work is a modeling/methods study using no human or animal data.

Data & code availability. All analysis code, constants, intermediate results and figures are released at https://github.com/Jefemaestro33/oxygenation-versus-tissue-composition-in-positronium-lifetime-pet-identifiability-limit (MIT for code; CC-BY-4.0 for text and figures, as specified in the repository license files).

Competing interests. The author declares no competing interests.

Funding. No specific funding was received.

Author contributions. E.D.P.Z. conceived the study, performed the analysis and wrote the manuscript. The analysis was implemented and cross-checked with AI-assisted pipelines under the author's direction; the AI systems are tools and are not authors.

Figure legends

Figure 1. Naive-reader apparent-pO_2 error (mmHg) when composition is ignored, over true pO_2 and lipid fraction. The read is usable only in a thin sliver at $f \approx 0$; a few percent lipid drives the apparent pO_2 hundreds of mmHg off.

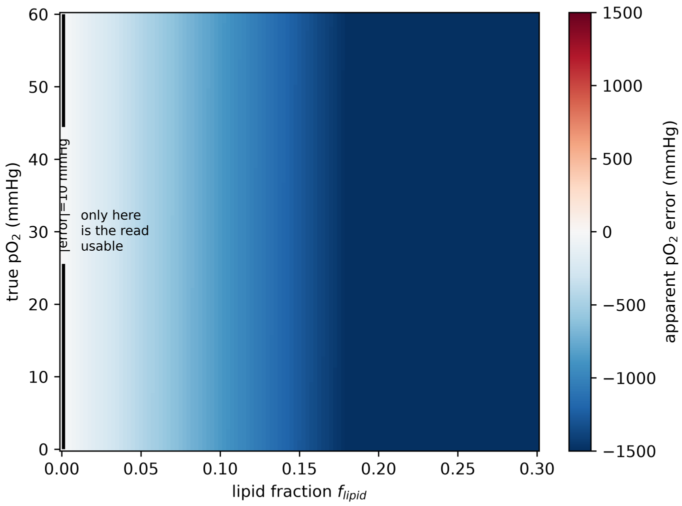


Figure 2. O_2 sensitivity $|d\tau_3/dpO_2|$ versus lipid fraction. The shaded band is the unmeasured lipid quenching constant ($\kappa_{\text{lipid}} = 0.1$–$1.0\times$ water); the approximately 1–10× net sensitivity spread is that unknown.

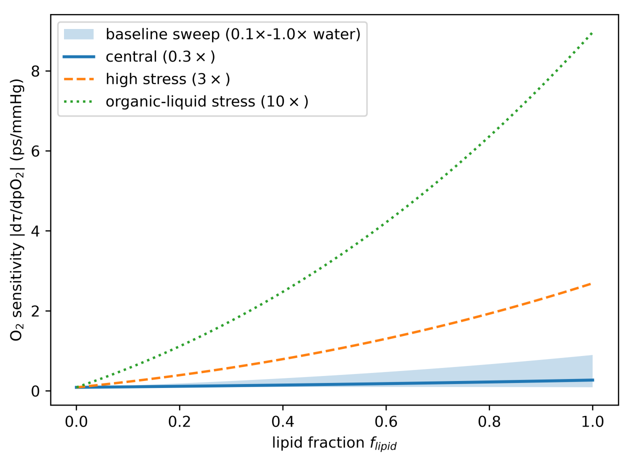


Figure 3. Cramér–Rao bound $\sigma(pO_2)$ versus total in-window counts for the full-spectrum (τ_3, I_3) fit; the lifetime-alone case is non-identifiable (∞). A 10 mmHg target needs order-10^9 in-window events, corresponding to order-10^8 o-Ps counts after the $N_3 = I_3(1 - \text{background})N_{\text{total}}$ conversion.

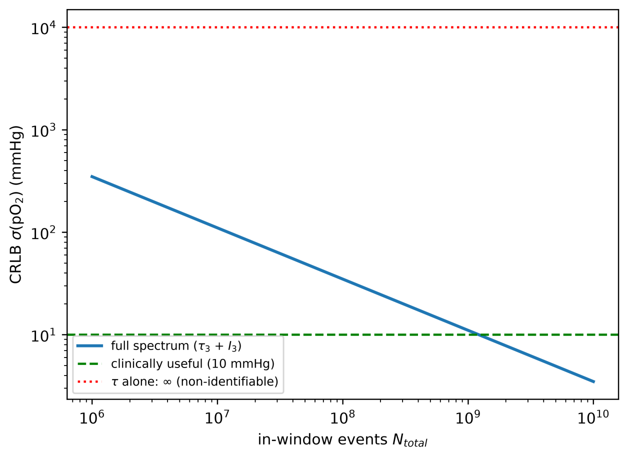


Figure 4. Why (τ_3, I_3) breaks the degeneracy. Iso-τ_3 contours run nearly along the ($\mathrm{pO_2}$,f) degeneracy direction (oxygen only slightly tilts τ_3); iso-I_3 contours pin composition. The crossing is shallow ⇒ identifiable but statistics-limited.

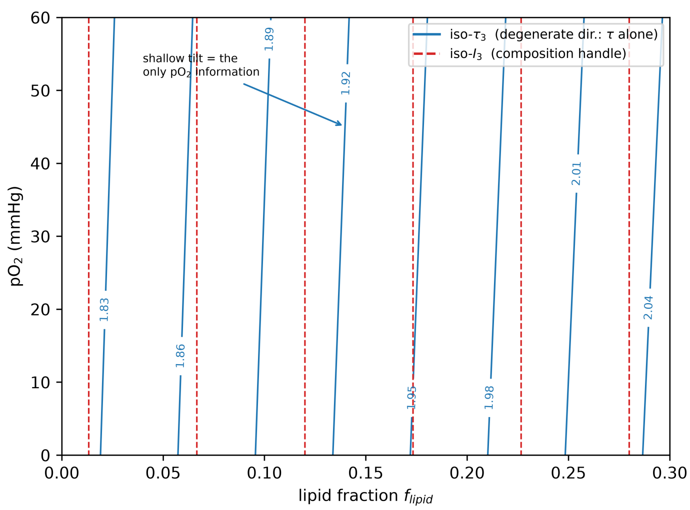


Figure 5. ROI volume required for $\sigma(pO_2) \leq 10$ mmHg versus lipid fraction, for three effective-yield isotope scenarios. Clinical separation needs litres of pooled signal; voxel-scale (about 0.06 mL) is far off-scale.

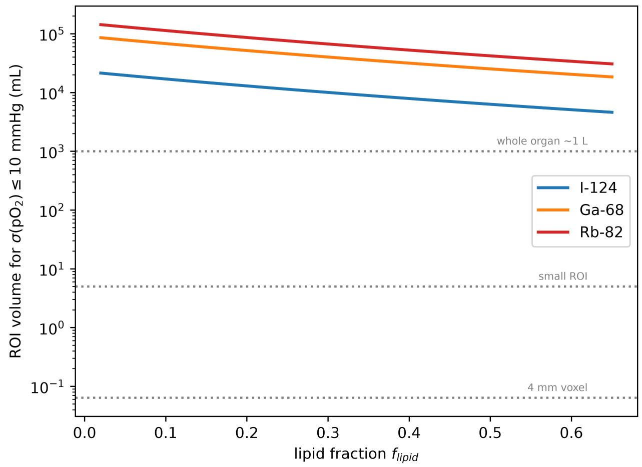
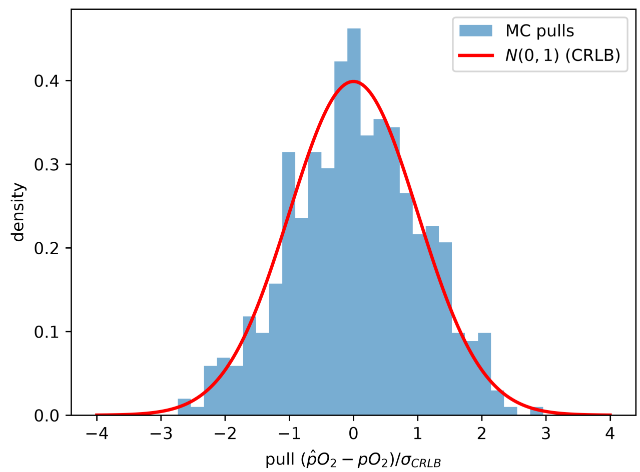


Figure 6. Virtual tumour (hypoxic core, normoxic rim, normal tissue): true pO_2 (left) and separability at a native 4 mm voxel (right) — impossible everywhere.

Supplementary figure

Figure S1. Parametric Monte-Carlo validation of the fixed-nuisance Poisson CRLB under the assumed spectrum model. Pulls of $\hat{p}O_2$ are close to $N(0, 1)$, showing that the maximum-likelihood estimator reaches the fixed-nuisance CRLB when the forward model is correctly specified.



Tables

Table 1. Forward-model constants (measured) and the two unmeasured unknowns.

Quantity	Value	Status	Source
τ_0 water	1.815 ns	measured	EJNMMI Phys 2024
τ_0 adipose	2.5–2.7 ns	measured	[7]
$\kappa(O_2)$ water	0.0204 $\mu mol^{-1}\mu s^{-1}L$	measured	[1,2]
O_2 solubility lipid:water	about 5×	measured	[9,10]
mechanism	spin (ortho→para) conversion	confirmed	[2]
κ_{lipid}	**unmeasured (swept 0.1–1.0×)**	**unknown**	§5
$\partial I_3/\partial$ **composition**	**unmeasured (assumed $\neq 0$)**	**unknown**	§5

Table 2. Cramér–Rao bound on pO_2 at $pO_2 = 20$ mmHg, $f = 0.10$, 10^5 o-Ps counts — robust to the noise model.

Noise model	$\sigma(pO_2)$ (mmHg)	ρ
Poisson, nuisances fixed (optimistic)	332	+0.70
Idealized analytic (field σ_τ)	570	+0.05
Poisson, per-spectrum nuisances profiled (realistic)	665	—
Poisson, all nuisances profiled (pessimistic)	758	—

References

1. Shibuya K, Saito H, Nishikido F, Takahashi M, Yamaya T. Oxygen sensing ability of positronium atom for tumor hypoxia imaging. *Commun Phys* 3, 173 (2020). doi:10.1038/s42005-020-00440-z
2. Stepanov PS, Selim FA, Stepanov SV, et al. Interaction of positronium with dissolved oxygen in liquids. *Phys Chem Chem Phys* 22, 5123 (2020). doi:10.1039/c9cp06105c
3. Moskal P, Kisielewska D, Curceanu C, et al. Feasibility study of the positronium imaging with the J-PET tomograph. *Phys Med Biol* 64, 055017 (2019). doi:10.1088/1361-6560/aafe20
4. Moskal P, Stępień EŁ. Positronium as a biomarker of hypoxia. *Bio-Algorithms Med-Syst* (2021). doi:10.1515/bams-2021-0189
5. Moskal P, Baran J, Bass S, et al. Positronium image of the human brain in vivo. *Sci Adv* 10, eadp2840 (2024). doi:10.1126/sciadv.adp2840
6. Mercolli L, Steinberger WM, Sari H, et al. In vivo positronium lifetime measurements with intravenous tracer administration and a long axial field-of-view PET/CT. *IEEE Trans Radiat Plasma Med Sci* (2026). doi:10.1109/TRPMS.2026.3687923
7. Avachat AV, et al. Ortho-positronium lifetime for soft-tissue classification. *Sci Rep* 14, 21155 (2024). doi:10.1038/s41598-024-71695-7 (arXiv:2409.13102)
8. Berens L, Hsu I, Chen CT, Halpern H, Kao CM. An analytic, moment-based method to estimate orthopositronium lifetimes in positron annihilation lifetime spectroscopy measurements. *Bio-Algorithms Med-Syst* 20(Spec. Issue), 40–48 (2024). doi:10.5604/01.3001.0054.9141
9. Battino R, Evans FD, Danforth WF. The solubilities of seven gases in olive oil. *J Am Oil Chem Soc* 45, 830 (1968). doi:10.1007/BF02540163
10. Roppongi T, Mizuno N, Miyagawa Y, Kobayashi T, Nakagawa K, Adachi S. Solubility and mass transfer coefficient of oxygen through gas- and water-lipid interfaces. *J Food Sci* 86, 867–873 (2021). doi:10.1111/1750-3841.15641
11. Mercolli L, Steinberger WM, Grundler PV, et al. First positronium lifetime imaging with scandium-44 on a long axial field-of-view PET/CT. arXiv:2506.13460 (2025).
12. Alkhorayef MA, Abuelhia EI, Chin MPW, Spyrou NM. Determination of the relative oxygenation of samples by ortho-positronium 3-gamma decay for future application in oncology. *J Radioanal Nucl Chem* 281, 171–174 (2009). doi:10.1007/s10967-009-0123-6
13. Alkhorayef M, Sulieman A, Alsager OA, Alrumayan F, Alkhomashi N. Investigation of using positronium and its annihilation for hypoxia PET imaging. *Radiat Phys Chem* 188, 109690 (2021). doi:10.1016/j.radphyschem.2021.109690
14. Sourbron SP, Buckley DL. Classic models for dynamic contrast-enhanced MRI. *NMR Biomed* 26, 1004–1027 (2013). doi:10.1002/nbm.2940
15. Köllner M, Wolfrum J. How many photons are necessary for fluorescence-

lifetime measurements? *Chem Phys Lett* 200, 199–204 (1992). doi:10.1016/0009-2614(92)87068-Z

16. Moskal P. Quantum entanglement degree, mean positronium lifetime, and the $3\gamma/2\gamma$ annihilation-rate ratio as novel PET biomarkers for hypoxia – concept, challenges, and predictions. *Bio-Algorithms Med-Syst* 22, 56 (2026). doi:10.5604/01.3001.0055.7461
17. Zermeño ED. Cardiac positronium lifetime in human PET: a reproducible right–left ventricular contrast that is not explained by blood oxygenation. *medRxiv* (2026). doi:10.64898/2026.06.14.26355630